

A Deterministic Defect Cocycle over the Quenched Residue–Height Spectral Shadow

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Remark (Status of these notes). This note builds on the pro-object $\mathfrak{Q}_{\text{Coll}}$ of the previous note (the quenched residue–height spectral shadow) and proposes the object that would make its central inverse arrow a theorem rather than a slogan. The construction is a reorganization of known finite-level structure in three places, marked as **[Inference]**, around one arrow that is genuinely open, marked as **[Speculation]**. The *Quenched Inverse Principle* is conjectural. Project-internal terms (H23a, the QSD machinery, the residue spaces Ω_Q , the tilted operators $\mathcal{L}_{s,Q,L,\chi}$) are carried over without restatement. Throughout, $e(t) = e^{2\pi it}$. Nothing here has been verified.

1 The obstacle that fixes the shape of the object

A single integer x_0 gives one deterministic valuation word $w(x_0) = (b_0, b_1, b_2, \dots)$, where $b_i = v_2(3x_i + 1)$. The annealed law is the pushforward of Haar measure on $\mathbb{Z}_2$ under $x \mapsto v_2(3x + 1)$. So “the orbit obeys the annealed law” means: the one prescribed word $w(x_0)$ equidistributes for the residue–height skew product. There is no ergodic theorem for a single prescribed non-generic point, and there cannot be: ergodicity yields almost-everywhere statements, which are silent about any individual orbit.

This dictates the shape of the missing object. It must be a *dichotomy with a closure*, not a measure-theoretic average: a pathwise quantity that is forced either to decay (the orbit equidistributes) or to localize as a finite, excludable obstruction. The quantity must reduce, at the trivial character and after ensemble averaging, to the annealed pressure $M(s)$ and the Cramér tilt; and its localization must land inside a fixed finite character sector, where H23a can act.

2 The object: a deterministic defect cocycle

Over each finite level (Q, L) , each tilt s , and each character $\chi \neq 1$ of Ω_Q , define the pathwise twisted Birkhoff functional

$$W_n(x; \chi, s) = \frac{1}{n} \sum_{i=0}^{n-1} \chi(x_i \bmod Q) 2^{s\Delta_{b_i}}, \quad \Delta_b = \log_2 3 - b.$$

The family of these functionals, fibered over $\mathfrak{Q}_{\text{Coll}}$, is the object; call it the *deterministic defect cocycle* $\mathfrak{D}_{\text{Coll}}$.

At $s = 0$ this is the residue Weyl sum: the orbit equidistributes modulo Q if and only if $W_n(x; \chi, 0) \rightarrow 0$ for every $\chi \neq 1$. The annealed pressure $M(s)$ and the critical tilt $p_b^{(1)} = 3 \cdot 4^{-b}$ are

the $\chi = \mathbf{1}$, ensemble-averaged shadow of the same functional. So $\mathfrak{D}_{\text{Coll}}$ is the quenched lift of the thermodynamic formalism: $M(s)$ is what W_n self-averages to when nothing obstructs.

The content of $\mathfrak{D}_{\text{Coll}}$ is a single identity, written in three languages, true at finite level and checkable:

$$\begin{aligned} \limsup_n |W_n(x; \chi, s)| &> 0 && \text{(equidistribution failure)} \\ \iff \chi \text{ is an approximate coboundary along the orbit} &&& \text{(affine / cohomological)} \\ \iff \mathcal{L}_{s,Q,L,\chi} \text{ has a near-peripheral eigenstate read off from } \mu_n^x &&& \text{(spectral).} \end{aligned}$$

The first equivalence is elementary: a twisted Birkhoff sum that does not decay telescopes precisely when the character is a near-coboundary along the orbit. The second is Perron–Frobenius duality: any weak-* limit of the empirical measures μ_n^x is invariant up to the telescoping error $\frac{1}{n}(\delta_{x_n} - \delta_{x_0}) = O(1/n)$, and a nonzero χ -component of an invariant measure is exactly a peripheral eigenstate of the twisted operator. **[Inference]**; this is standard for finite-state transfer operators, transcribed onto the operators $\mathcal{L}_{s,Q,L,\chi}$.

3 Why the annealed side has a gap, explicitly

At finite level the gap is computable. Use the exact affine cocycle

$$x_n = \frac{3^n x_0 + \sum_{j=0}^{n-1} 3^{n-1-j} 2^{B_j}}{2^{B_n}}, \quad B_j = \sum_{i < j} b_i.$$

Reduce modulo 3^r . For $n \geq r$ the 3^n term vanishes and only the last r valuations survive:

$$x_n \equiv \sum_{k=1}^r 3^{k-1} 2^{-(b_{n-1} + \dots + b_{n-k})} \pmod{3^r}.$$

Since 2 is a primitive root modulo 3^r (order $2 \cdot 3^{r-1}$), the character value $\chi_3(x_n)$ is a sliding-window function of the partial sums $B_i \pmod{2 \cdot 3^{r-1}}$. The single-symbol Fourier transform of the annealed law is

$$\hat{p}(\xi) = \sum_{b \geq 1} 2^{-b} e(b\xi) = \frac{e(\xi)}{2 - e(\xi)}, \quad |\hat{p}(\xi)| = \frac{1}{\sqrt{5 - 4 \cos 2\pi\xi}} < 1 \quad (\xi \notin \mathbb{Z}).$$

The frequencies attached to a nontrivial χ are $\xi = k/(2 \cdot 3^{r-1})$, all nonintegral, so $|\hat{p}(\xi)| < 1$ at each. To leading order in the i.i.d. model,

$$\rho(\mathcal{L}_{0,Q,L,\chi}) = \max_{\xi} |\hat{p}(\xi_{\chi})| < 1.$$

[Inference]; this is the leading-order i.i.d. value, not the exact spectral radius of the coupled finite operator, but it pins the mechanism and the gap constant κ_Q . That explicit strict inequality is H23a made arithmetic: nontrivial characters decorrelate because $5 - 4 \cos 2\pi\xi > 1$ off the integers.

4 The Quenched Inverse Principle

Quenched Inverse Principle (conjectural)

Let x_0 have an infinite forward orbit, and suppose it violates the annealed law persistently: for some $\chi \neq 1$ on some Ω_Q , some tilt s , and some $\delta > 0$,

$$|W_n(x_0; \chi, s)| \geq \delta \quad \text{along a positive-density set of } n.$$

(This covers the three concrete failures: a sub-equilibrium tail $\bar{b}_n < \log_2 3$; persistent high-ascent shadowing $D(\mu_n \| p^{(1)}) \rightarrow 0$; and residue bias.) Then there exist a finite pair (Q, L) and a χ -sector eigenstate of $\mathcal{L}_{s,Q,L,\chi}$ with eigenvalue

$$|\lambda| \geq \rho(\mathcal{L}_{s,Q,L,0}) - o(1),$$

exhibited by the orbit's own empirical measure. Conjoined with the gap $\rho(\mathcal{L}_{s,Q,L,\chi}) \leq (1 - \kappa_Q)\rho(\mathcal{L}_{s,Q,L,0})$, this is a contradiction. Hence every integer orbit equidistributes and obeys the annealed law.

The contradiction chain is the one stated before: bad orbit $\Rightarrow$ finite spectral/coboundary shadow $\Rightarrow$ violation of H23a. The role of $\mathfrak{D}_{\text{Coll}}$ is to make the first arrow a theorem-shaped statement.

5 The keystone, where the real theorem lives

Everything above is reorganization. The single arrow that is not a transcription of known structure is

$$\begin{aligned} & \text{persistent quenched bias along } w(x_0) \\ \implies & \text{ a **finite-level** approximate eigenfunction at the **top** eigenvalue.} \end{aligned}$$

The empirical-measure argument supplies an invariant functional, but invariance together with nonzero χ -mass does not by itself localize the obstruction at bounded Q with eigenvalue pushed all the way to the principal one. Two leaks must be closed, and naming them is the useful part.

Finiteness leak. The limiting obstruction could in principle live at $Q \rightarrow \infty$, an obstruction that recedes to the boundary and never appears at any fixed modulus. Closing this requires an effective, uniform gap,

$$\kappa_Q \gtrsim \frac{1}{\text{polylog}(Q)},$$

decaying more slowly than the entropy production of the bias. That quantitative H23a is a hypothesis, not a consequence; it is the precise point where “annealed-to-quenched” becomes a rate inequality.

Circularity leak. Reading a top-eigenvalue eigenfunction off μ_n^x presumes the orbit's residue frequencies are already near-stationary, which is what one is trying to prove. Breaking this requires an input that does not come from the orbit's own statistics.

My best candidate for the circularity-breaking input, pressed as far as I will commit, is a regeneration-plus-bilinear engine resting on the multiplicative independence of 2 and 3. The valuation word is 2-adic data; the residue modulo 3^r is 3-adic data; the map $3x + 1$ couples

them. Open the Weyl sum bilinearly across the two scales and apply a Bourgain–Sarnak–Ziegler orthogonality step over the 3-adic frequencies, using that the 2-adic increments at distinct 3-adic frequencies are jointly nonresonant ($|\hat{p}(\xi)| < 1$ at every $\xi \in \frac{1}{2 \cdot 3^r - 1} \mathbb{Z} \setminus \mathbb{Z}$). The regeneration times are the indices where the height walk hits fresh record lows, which re-uniformize the residue; between regenerations the increments are independent enough for a quenched Borel–Cantelli, summable because the bad-stretch probability is $\leq e^{-c\ell}$ on a stretch of length ℓ . **[Speculation]**, on the structure of a closure; no such argument exists, and a Bourgain–Sarnak–Ziegler step on a noninvertible monoid with arithmetically determined inputs is exactly the unsolved part.

The honest counterweight: that keystone is morally a normality / disjointness statement for the 2-adic word of the integers, and it is Sarnak-hard. The deterministic word $w(x_0)$ is zero-entropy data attached to one integer, and demanding that it decorrelate from the 3-adic character system for every x_0 has the same flavor of difficulty as Möbius disjointness for low-complexity sequences. So $\mathfrak{D}_{\text{Coll}}$ does not dissolve Collatz; it relocates the difficulty into one named, falsifiable lemma and gives that lemma an exact finite-level target.

6 Making “annealed-to-quenched” literal

The statistical-mechanics gloss sharpens the prediction rather than decorating it. The annealed pressure is $\log M(s)$. The quenched pressure along an orbit is

$$\Lambda^{\text{que}}(s; x_0) = \lim_n \frac{1}{n} \log \sum_{w_n} (\text{tilted weights along the orbit}).$$

Quenched equal to annealed is self-averaging of the free energy. A bad orbit is a self-averaging failure, the Collatz analogue of replica-symmetry breaking, and the inverse principle asserts that every such failure is carried by a finite character sector. So the conjecture reads, concretely:

The quenched Collatz pressure self-averages with no replica-symmetry breaking, and any RSB component is a peripheral eigenstate of some $\mathcal{L}_{s,Q,L,\chi}$ with $\chi \neq 1$.

[Inference] for the framing; **[Speculation]** that no-RSB holds.

7 Three falsifiable consequences computable now

These would weaken or strengthen the object without waiting on the keystone.

1. For each r , compute the exact $\rho(\mathcal{L}_{0,3^r,L,\chi})$ and check that it tracks

$$\max_{\xi} |\hat{p}(\xi)| = \max_k (5 - 4 \cos(2\pi k / (2 \cdot 3^{r-1})))^{-1/2}.$$

If the coupled operator’s gap is parametrically smaller than the i.i.d. value, the finiteness leak is real, and the computation returns the true rate of κ_Q .

2. Along long orbits, compute the windowed Weyl sums $W_n(x; \chi, 0)$ at the resonant frequencies $k/(2 \cdot 3^{r-1})$. The prediction is decay at rate $|\hat{p}(\xi)|^n$ with no persistent plateau. A plateau at any fixed χ for any orbit is the spectral shadow made visible, and a direct test of the circularity leak.

3. At $s = 1$, compare $\rho(\mathcal{L}_{1,3^r,L,\chi})$ with $\rho(\mathcal{L}_{1,3^r,L,0})$. A nontrivial gap there is the statement that a high ascent cannot also hold a residue resonance, which is the joint annealed-to-quenched obstruction for critical excursions.

Summary

Component	Status
$W_n(x; \chi, s)$, the defect cocycle $\mathfrak{D}_{\text{Coll}}$ three-language equivalence at fixed (Q, L)	quenched lift of the thermodynamic formalism finite, checkable [Inference]
$\rho(\mathcal{L}_{0,Q,L,\chi}) = \max_{\xi} \hat{p}(\xi_{\chi}) < 1$ Quenched Inverse Principle	H23a made arithmetic [Inference] the missing theorem (conjectural)
keystone: bias $\Rightarrow$ top finite eigenstate no-RSB self-averaging	one open lemma [Speculation] ; Sarnak-hard equivalent reframing [Speculation]

The object relocates the foundational difficulty rather than removing it: $\mathfrak{D}_{\text{Coll}}$ turns “the orbit should behave randomly” into one falsifiable lemma, namely that a persistent quenched bias must surface as a top-eigenvalue eigenstate in a finite character sector, where the arithmetic gap $|\hat{p}(\xi)| < 1$ already forbids it.